TAJWARUL HASAN CHOWDHURY

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LinkedIn | GitHub | Portfolio

SUMMARY

Software Engineer building full-stack and AI-powered applications with TypeScript, React, Next.js, Node.js, PostgreSQL, and MongoDB. Experienced in backend and REST API design, authentication and security, asynchronous background processing, and LLM integration, including RAG pipelines with LangChain. Codeforces Pupil with 1,300+ problems solved.

EDUCATION

Sylhet Engineering College | B.Sc. in Computer Science and Engineering | CGPA: 3.69 / 4.00 **Feb 2022 – Aug 2026**

TECHNICAL SKILLS

Languages: C++, JavaScript, TypeScript
Frontend: React, Next.js, HTML5, CSS3, Tailwind CSS, shadcn/ui
Backend: Node.js, Express.js, REST APIs, Zod (schema validation)
Databases & ORM: PostgreSQL, MongoDB, Drizzle ORM, Mongoose, Neon
Authentication & Security: JWT, Better Auth, Firebase Authentication, Role-Based Access Control (RBAC)
AI & LLM: LangChain, Retrieval-Augmented Generation (RAG), Qdrant, LLM API Integration (Groq, Google Gemini)
Deployment & Tools: Git, GitHub, Vercel, Render, Cloudinary
Core CS: Data Structures & Algorithms, Object-Oriented Programming (OOP), Operating Systems, Computer Networks, Database Management Systems (DBMS)

ACHIEVEMENTS

- Codeforces max rating 1267 (Pupil), handle: tajwarul
- Solved 1,300+ algorithmic and data structure problems across multiple online programming platforms.
- Participated in 40+ competitive programming contests.

PROJECTS

HireFlow | AI-Assisted Hiring Platform
React, *Express 5*, *MongoDB (Mongoose)*, *Groq*, *Puppeteer*, *Firebase Auth*, *Cloudinary* | GitHub · Live

- Built a two-sided hiring platform for candidates and recruiters, featuring AI-generated resumes, job-specific readiness reports, AI-ranked applicants, and hiring pipeline management.
- Designed AI workflows with **Groq** + **Zod**-validated structured outputs and **asynchronous processing** for application scoring, PDF generation, and file uploads.
- Implemented **RBAC**, company-scoped authorization, **JWT** authentication, Google Sign-In, recruiter invitations, **rate limiting**, and **MongoDB** indexing for secure and scalable data access.

Scam Scanner | AI Scam Evidence Investigator
Next.js 16, *TypeScript*, *PostgreSQL (Neon)*, *Drizzle ORM*, *BullMQ*, *Redis*, *Gemini*, *Groq* | GitHub · Live

- Built an AI-powered scam analysis tool that processes screenshots, PDFs, DOCX files, and text to generate risk reports, detect contradictions, and provide verification steps.
- Engineered an asynchronous **BullMQ** pipeline using **Gemini** multimodal extraction with **Groq** fallback, combined with rule-based signals and URL analysis using RDAP, Safe Browsing, and homograph detection.
- Secured the pipeline with file-type validation, DOCX zip-bomb detection, **prompt-injection defenses**, **Zod** validation, Better Auth, and Upstash Redis **rate limiting**.

SEC Library Management System | University Library Platform
React, *Express 5*, *MongoDB (Mongoose)*, *LangChain*, *Qdrant*, *Groq*, *SSLCommerz* | Student Portal · Admin Portal · GitHub

- Built student and admin React portals for catalog search, borrowing, reservations, waitlists, fines, and paper approvals.
- Integrated Firebase Google Sign-In and **SSLCommerz** payments with server-side validation and **idempotent** transactions.
- Developed a **RAG** catalog chatbot using LangChain, Gemini embeddings, **Qdrant**, and Groq with conversation memory.